Muscat 2040: Growth & Infrastructure Challenge

Executive Summary

This submission projects **Muscat Governorate's population to 2040** under three growth scenarios and translates population expansion into **infrastructure demand pressures** for:

- Healthcare capacity & Housing demand

All estimates are produced using **transparent and reproducible calculations** derived from the provided workbook, with assumptions fully documented in the **Technical Appendix**.

Headline Results (2040 Population)

Using the workbook's Muscat population baseline and the scenario assumptions documented in the Technical Appendix:

Scenario	Population (2040)
Low Growth	**2.82 million**
Base Growth	**3.08 million**
High Growth	**3.41 million**

These projections imply **continued strong urban expansion**, with Muscat adding **~1–1.6 million additional residents** relative to early-2020s levels depending on the scenario.

Healthcare Risk: Hospital Bed Demand vs Current Capacity

Healthcare demand is estimated using a planning benchmark of:

3 hospital beds per 1,000 population

Current capacity baseline:

MOH Muscat beds (2023): 1,608

Projected Demand vs Capacity (2040)

Scenario	Bed Demand	Capacity	Gap	Stress Index
Low	**8,464**	1,608	**6,856**	**5.26**
Base	**9,230**	1,608	**7,622**	**5.74**
High	**10,238**	1,608	**8,630**	**6.37**

Stress Index = Demand ÷ Capacity

Breakpoint Year

Under the 3 beds per 1,000 planning benchmark:

Demand already exceeds capacity in 2023 across all scenarios.

This indicates that **Muscat's healthcare system is already operating below the benchmark planning level**, and population growth will widen the gap without capacity expansion.

Housing Risk: Households Required (Demand-Side)

Housing demand is estimated using:

$$Households = \frac{Population}{Household\ Size}$$

Using a household-size proxy of:

6.27 persons per household

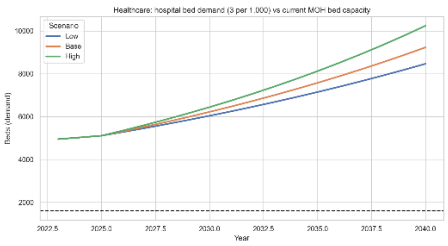
Derived from the workbook's **Oman household-size series**.

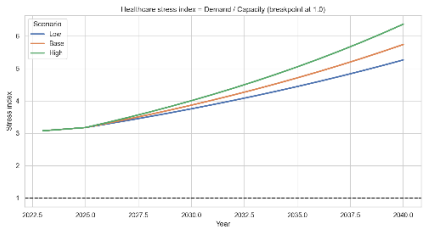
Estimated Housing Demand by 2040

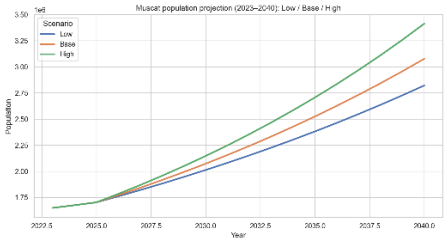
Scenario	Households Required
Low	~449,899 households
Base	~490,599 households
High	~544,198 households

Because the workbook does not include **Muscat housing stock**, this analysis reports **demand-side household requirements only**. A full supply gap calculation would require housing inventory data.

Key Charts (Submission Figures)







Implications for Decision-Makers

1. Muscat's 2040 planning population range is ~2.8M–3.4M

Even the **low-growth scenario** implies major increases in service demand relative to the early 2020s.

Urban planning, utilities, healthcare, and transport systems will need to scale accordingly.

2. Healthcare capacity is the most immediate infrastructure bottleneck

Under the planning benchmark of 3 beds per 1,000 residents:

- Hospital bed demand could reach **5–6× current MOH capacity by 2040**.
- Capacity expansion, service redesign, and private sector participation will be critical.

3. Housing delivery requirements are substantial

Muscat may require approximately:

450,000–544,000 housing units by 2040

Meeting this demand will require coordinated policies for:

- Land supply
- Residential construction
- Infrastructure provision
- Urban density management

Muscat's housing demand also implies a sustained annual construction requirement.
Assuming the current housing stock supports approximately **~280,000 households** (illustrative estimate), the **base scenario** implies the need for roughly:

~10,000–12,000 new housing units per year between 2024 and 2040.

This highlights the need for **consistent residential development, serviced land supply, and supporting infrastructure** to accommodate projected population growth.

Three Policy-Ready Recommendations

1. Implement a staged healthcare capacity expansion strategy (2026–2040)

Adopt the **Stress Index (Demand / Capacity)** as a governance indicator.

Define clear thresholds:

Stress Level	Action
Green (<1.5)	Monitor
Amber (1.5–3)	Plan expansion
Red (>3)	Accelerate capital investment

A rolling **5-year infrastructure investment pipeline** can then be triggered automatically.

2. Shift from "beds-only" planning to integrated healthcare delivery

Hospital beds should not be the sole response to demand.

Complementary solutions include:

- Day surgery expansion
- Step-down and rehabilitation facilities
- Telemedicine and remote care
- Referral management systems
- Public–private healthcare coordination

These measures can reduce pressure on inpatient hospital capacity.

3. Establish a coordinated housing delivery program

Translate population scenarios into **annual housing targets** and align them with:

- serviced land supply and utilities expansion
- transport corridors and urban growth nodes

Integrated planning will reduce the risk of **unplanned sprawl and infrastructure deficits**.